

## CIRIK-RABAT, Kazakhstan

WMO: 380490

Lat: 44.067N

Lon: 62.900E

Elev: 88

StdP: 100.27

Time Zone: 5.00 (E05)

Period: 82-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 2             | -22.0      | -18.8 | -25.9                         | 0.4 | -20.5 | -22.9 | 0.5 | -18.5 | 12.2                  | -7.2 | 10.3 | -10.4 | 1.2                   | 20   | 0.512 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.1                   | 39.7            | 22.6 | 38.0 | 21.6 | 36.4 | 21.1 | 26.1                | 35.6 | 24.3 | 34.0 | 22.8 | 33.1 | 2.7                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.0                            | 17.9 | 33.9 | 20.9 | 15.8 | 31.4 | 19.1 | 14.0 | 30.3 | 81.5          | 35.5 | 73.7 | 33.9 | 67.9 | 33.1 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.3           | 8.0  | 6.9 | -23.3                      | 43.1 | 4.8                | 2.1 | -26.8                                              | 44.6 | -29.6      | 45.8 | -32.3      | 47.0 | -35.9      | 48.5 |    |    |
| (5)               |      |     | -23.6                      | 26.3 | 4.8                | 3.3 | -27.0                                              | 28.7 | -29.8      | 30.6 | -32.5      | 32.5 | -36.0      | 34.9 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.0   | -5.6  | -5.7  | 2.1   | 13.2  | 20.0  | 26.1  | 28.9  | 25.8  | 18.7  | 10.1  | 1.5   | -3.8  |  |  |
|                                                                        | DBStd       | 13.40  | 6.05  | 6.62  | 5.67  | 5.11  | 4.38  | 3.82  | 3.10  | 3.39  | 4.05  | 5.09  | 5.94  | 6.16  |  |  |
|                                                                        | HDD10.0     | 1949   | 482   | 441   | 252   | 26    | 2     | 0     | 0     | 0     | 1     | 59    | 258   | 428   |  |  |
|                                                                        | HDD18.3     | 3611   | 741   | 674   | 503   | 166   | 33    | 2     | 0     | 1     | 43    | 258   | 503   | 686   |  |  |
|                                                                        | CDD10.0     | 2326   | 0     | 0     | 7     | 122   | 312   | 482   | 585   | 489   | 262   | 63    | 4     | 0     |  |  |
|                                                                        | CDD18.3     | 946    | 0     | 0     | 0     | 12    | 85    | 234   | 327   | 231   | 54    | 3     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 13367  | 0     | 0     | 1     | 164   | 1093  | 3331  | 4926  | 3069  | 721   | 63    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 7370   | 0     | 0     | 0     | 43    | 444   | 1886  | 3034  | 1672  | 276   | 14    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.1    | 1.8   | 2.4   | 2.6   | 2.7   | 2.2   | 2.4   | 2.3   | 2.1   | 1.9   | 1.7   | 1.9   | 1.7   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 179    | 26    | 26    | 27    | 18    | 16    | 8     | 3     | 3     | 1     | 7     | 20    | 22    |  |  |
|                                                                        | PrecMax     | 294    | 65    | 62    | 62    | 57    | 58    | 36    | 18    | 21    | 8     | 26    | 58    | 65    |  |  |
|                                                                        | PrecMin     | 98     | 5     | 6     | 7     | 3     | 2     | 1     | 0     | 0     | 0     | 1     | 3     | 2     |  |  |
|                                                                        | PrecStd     | 52     | 13    | 15    | 12    | 12    | 14    | 9     | 4     | 5     | 2     | 6     | 14    | 13    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 9.9   | 12.4  | 20.4  | 31.0  | 35.5  | 40.9  | 42.4  | 39.9  | 35.3  | 28.5  | 19.5  | 10.8  |  |  |
|                                                                        |             | MCWB   | 5.7   | 7.4   | 11.7  | 18.5  | 19.6  | 24.1  | 24.2  | 23.5  | 19.6  | 16.5  | 11.5  | 6.2   |  |  |
|                                                                        | 2%          | DB     | 5.6   | 7.0   | 16.6  | 27.4  | 33.0  | 38.7  | 40.6  | 37.8  | 32.1  | 24.6  | 14.9  | 7.1   |  |  |
|                                                                        |             | MCWB   | 3.1   | 4.8   | 9.7   | 16.5  | 18.7  | 21.6  | 22.6  | 21.6  | 18.1  | 14.2  | 8.8   | 4.0   |  |  |
|                                                                        | 5%          | DB     | 2.8   | 4.1   | 13.5  | 25.0  | 31.0  | 36.7  | 38.8  | 36.0  | 30.0  | 22.0  | 12.1  | 4.7   |  |  |
|                                                                        |             | MCWB   | 1.0   | 1.8   | 8.3   | 15.0  | 17.6  | 21.1  | 21.9  | 20.8  | 17.2  | 12.8  | 7.4   | 2.5   |  |  |
|                                                                        | 10%         | DB     | 1.4   | 2.1   | 10.3  | 22.5  | 29.1  | 34.8  | 37.1  | 34.2  | 27.6  | 19.1  | 9.4   | 3.2   |  |  |
|                                                                        |             | MCWB   | -0.1  | 0.3   | 6.4   | 13.7  | 17.1  | 20.6  | 21.6  | 19.9  | 16.0  | 11.0  | 6.0   | 1.4   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 5.9   | 8.5   | 12.1  | 19.8  | 22.8  | 27.1  | 29.5  | 27.7  | 22.9  | 17.4  | 12.2  | 6.3   |  |  |
|                                                                        |             | MCDB   | 9.1   | 11.7  | 19.8  | 29.2  | 30.7  | 37.7  | 39.9  | 36.7  | 32.0  | 26.3  | 17.9  | 9.6   |  |  |
|                                                                        | 2%          | WB     | 3.1   | 4.9   | 10.5  | 17.4  | 20.8  | 24.6  | 26.7  | 24.8  | 20.4  | 15.2  | 9.8   | 4.3   |  |  |
|                                                                        |             | MCDB   | 5.5   | 6.7   | 15.6  | 25.8  | 29.9  | 34.4  | 36.2  | 34.6  | 28.7  | 23.7  | 13.5  | 6.8   |  |  |
|                                                                        | 5%          | WB     | 1.4   | 2.1   | 8.7   | 15.9  | 19.1  | 23.2  | 24.9  | 23.0  | 18.1  | 13.2  | 8.0   | 2.9   |  |  |
|                                                                        |             | MCDB   | 2.6   | 3.9   | 13.4  | 23.8  | 28.8  | 33.6  | 34.5  | 32.8  | 27.6  | 20.5  | 11.5  | 4.6   |  |  |
|                                                                        | 10%         | WB     | 0.1   | 0.4   | 6.6   | 14.2  | 17.7  | 21.8  | 23.3  | 21.2  | 16.5  | 11.4  | 6.1   | 1.6   |  |  |
|                                                                        |             | MCDB   | 1.2   | 2.0   | 10.1  | 21.5  | 27.8  | 32.7  | 33.6  | 31.5  | 26.3  | 18.5  | 9.2   | 3.0   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 7.3   | 8.6   | 9.3   | 12.8  | 13.8  | 14.3  | 14.1  | 14.1  | 14.6  | 13.4  | 9.2   | 7.2   |  |  |
|                                                                        |             | MCDBR  | 7.7   | 9.5   | 14.3  | 15.9  | 16.4  | 16.2  | 16.3  | 16.5  | 17.4  | 16.9  | 13.5  | 9.3   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.0   | 7.1   | 8.6   | 8.9   | 8.1   | 8.0   | 7.5   | 8.1   | 9.3   | 10.0  | 8.9   | 7.3   |  |  |
|                                                                        |             | MCWBR  | 7.1   | 9.5   | 13.3  | 15.0  | 15.4  | 14.8  | 14.9  | 15.3  | 16.0  | 16.3  | 12.3  | 8.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.362 | 0.391 | 0.449 | 0.493 | 0.495 | 0.477 | 0.456 | 0.440 | 0.409 | 0.405 | 0.385 | 0.365 |  |  |
|                                                                        | taud        |        | 2.075 | 2.021 | 1.917 | 1.844 | 1.888 | 1.974 | 2.057 | 2.073 | 2.105 | 2.055 | 2.079 | 2.071 |  |  |
|                                                                        | Ebn at Noon |        | 713   | 765   | 769   | 771   | 786   | 801   | 814   | 812   | 804   | 739   | 677   | 660   |  |  |
|                                                                        | Edh at Noon |        | 105   | 132   | 167   | 195   | 193   | 178   | 161   | 153   | 137   | 126   | 104   | 96    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.49  | 2.57  | 3.94  | 5.45  | 6.88  | 7.55  | 7.36  | 6.70  | 5.30  | 3.46  | 2.00  | 1.32  |  |  |
|                                                                        | RadStd      |        | 0.22  | 0.39  | 0.35  | 0.41  | 0.38  | 0.33  | 0.27  | 0.22  | 0.19  | 0.20  | 0.22  | 0.18  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (3 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | -1.27 | N/A | N/A         | N/A     | +79     |         |

Nomenclature: See separate page